01276314 Microcontroller Interfacing

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Instructor

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Course Description

Microcontroller Interfacing

- Microcontroller Architecture
- Digital Input and Output
- Serial Communications
- Analog-to-Digital Conversion
- Digital-to-Analog Conversion
- Timers
- Interrupt Handling
- Sensors and Actuators
- Displays
- Memory Technology
- Microcontroller Programming and Development

Program Learning Outcome

1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

3) an ability to communicate effectively with a range of audiences.

4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

6) an ability to develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions.

7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Curriculum Mapping

● Major Responsibility ○ Minor Responsibility

Courses	Morality and Ethics							Knowledge								Cognitive skills				Interpersonal skills and responsibility						Analytical and communication skills			
	1	2	3	4	5	6	7	1	2	3	4	5	6	7	8	1	2	3	4	1	2	3	4	5	6	1	2	3	4
Mathematics and Basic Sciences																													
01006723 General Physics 1		○						●								●	○		○								●		
01006724 General Physics Laboratory 1		○						●								●	○		○								●		
01006725 General Physics 2		○						●								●	○		○								●		
01006726 General Physics Laboratory 2		○						●								●	○		○								●		
01006710 Intro to Calculus		○						●								●	○		○								●		
01006711 Advanced Calculus		○						●								●	○		○								●		
01006717 Differential Equations		○						●								●	○		○								●		
01276206 Probability and Random Variables		○						●								●	○		○								●		
01276207 Introduction to Statistics		○						●								●	○		○							●	●		
01006716 Linear Algebra		○						●								●	○		○								●		
01276209 Discrete Mathematics		○						●								●	○		○								●		
Computer Hardware and Architecture																													
01276111 Circuits and Electronics		○						●	●							○	○		●				●			●			
01276112 Digital System Fundamentals		○						●	●	●						○	○		●				●			●			
01276213 Computer Architecture and Organization		○						●	●							○			●				●			●			
01276314 Microcontroller Interfacing		○						●	●	●						○			●				●			●			
Systems Infrastructure																													
01276121 Computer Programming		○						●	●					○		○	○									●			
01276222 Data Structure and Algorithms		○						●	●					○		○	○									●			
01276223 Computer Networks		○						●	●	●						○			●				●			●			
01276324 Operating Systems		○						●	●							○			●				●			●			
01276325 Information and Computer Security		○						●	●	●						○		○	●				●			●			
01276326 Theory of Computation		○						●								○			○								●		

Morality and Ethics

2. Have discipline, punctuality, and self and social responsibility

Knowledge

1. Have acquired knowledge and understandings of important principles and theories of the subjects in the field of study.

2. Be able to analyze problems, understand and explain computing requirements, as well as be able to apply knowledge, skills, and the usage of suitable tools for solving problems.

3. Be able to analyze, design, deploy, improve, and/or evaluate computing systems and their components in accordance with the requirements.

Cognitive

1. Have learned to think analytically and systematically.

4. Be able to apply knowledge and skills in solving computing related problems properly.

Interpersonal Skills and Responsibility

4. Be responsible for own actions and for the team.

Analytical and Communication

1. Have acquired skills in utilizing the existing tools that are essential in computer related work.

Course Learning Outcomes

CLO-1. Explain embedded systems concept.

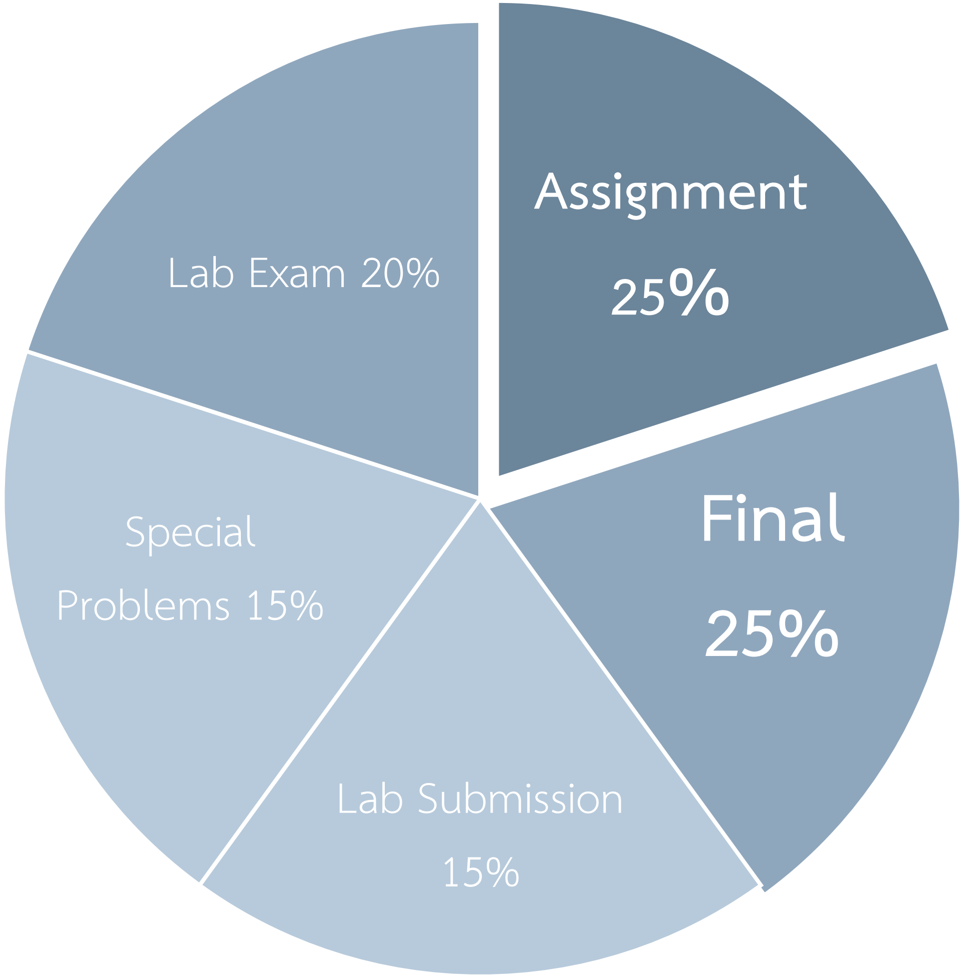
CLO-2. Understand the principle of microcontroller.

CLO-3. Apply microcontroller for several types of application.

CLO-4. Use external sensors and actuators with microcontroller.

CLO-5. Develop microcontroller application on STM32 platform.

Grading : Norm-Referenced Assessment



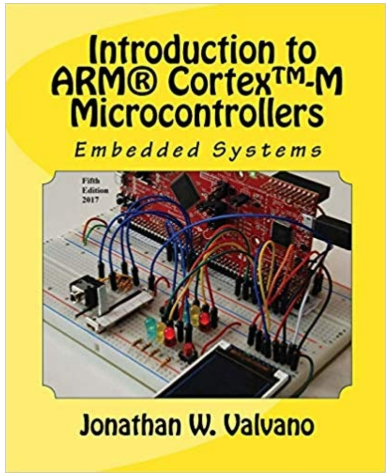
Reference

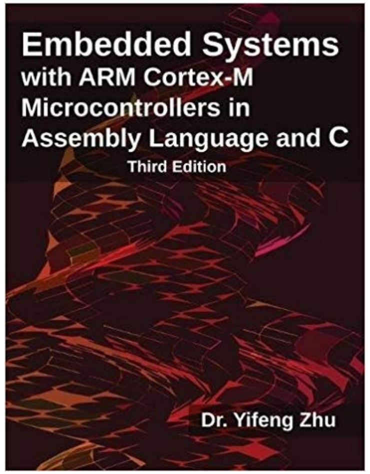
[1] Jonathan W. Valvano, **Embedded Systems: Introduction to Arm® Cortex™-M Microcontrollers, Vol 1, 2nd Edition,** 2012

[2] Yifeng Zhu, **Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C,** 3rd Edition, 2017

[3] www.arm.com

[4] www.st.com





Other Resources

STMicroelectronics
29.2K subscribers

https://www.youtube.com/user/STonlineMedia

Web learning
6.12K subscribers

https://www.youtube.com/channel/UCjPRuknkFjqM6UxxTbCQJ5Q

Controllers Tech
21K subscribers

https://www.youtube.com/c/ControllersTech

hackster.io
AN AVNET COMMUNITY

https://www.hackster.io/

HACKADAY.IO

https://hackaday.io/

Other Resources

Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C

10,047 subscribers

https://www.youtube.com/channel/UCY0sQ9hpSR6yZobt1qOv6DA

MYaqoobEmbedded

6,685 subscribers

https://www.youtube.com/channel/UC-CuJ6qKst9-8Z-EXjoYK3Q

Quantum Leaps, LLC

30,489 subscribers

https://www.youtube.com/channel/UCMGXFEew8I6gzjg3tWen4Gw

Syllabus

- Introduction to microprocessor, microcontroller and embedded systems
- Architecture and organization of Cortex M
- Peripherals
  - GPIO, Interrupt, Timer, ADC, DAC
  - UART, I2C, SPI
  - LCD, Touch Sensor
- Debugging

TENTATIVE SYLLABUS

Week	Lecture	Lab
1	Introduction to Embedded Systems	Getting Started
2	Cortex Architecture	LED
3	General Purpose Input Output	General Purpose Input Output
4	Embedded C	Receiver Transmitter (UART)
5	Receiver Transmitter (UART)	Nested-Vector Interrupt Controller
6	Interrupt	Analog to Digital Converter
7	Analog to Digital Converter	Timer
8	Timer	Pulse-width Modulation
9	Pulse-width Modulation	Liquid Crystal Display & Touch Sensor
10	Serial Peripheral Interface	Assignment
11	Liquid Crystal Display & Touch Sensor	Assignment
12	Assignment	Assignment
13	Assignment	Assignment
14	Assignment	Assignment
15	Assignment	Assignment

Past Assignments

https://bit.ly/2JR0zNs